

# 1 Notation

## 1.1 Weyl Spinors

There exist four different types of Weyl spinors transforming in four different but pairwise equivalent two-dimensional representations of  $\text{SL}(2, \mathbb{C})$ . For future purpose it is enough to consider only the covariant representation transforming w.r.t. to  $A$  and the conjugate covariant representation transforming with  $(A^{-1})^\dagger$ . These two representations are inequivalent and the two missing equivalent can be obtained by the relation  $\varepsilon A \varepsilon^{-1} = (A^{-1})^T$  with  $\varepsilon$  given in (1.8). For more detailed derivations and explanations of the following two chapters see [5, p. 30–32] and [7, chap. 34 & 35].

Two spinors  $\xi_\alpha(\mathbf{p})$  and  $\bar{\chi}^{\dot{\alpha}}(\mathbf{p})$  transforming w.r.t. the just mentioned representations transform like:

$$\begin{aligned}\xi_\alpha(\mathbf{p}) &\rightarrow \xi'_\alpha(\mathbf{p}) = A_\alpha{}^\beta \xi_\beta(\Lambda^{-1}(A)\mathbf{p}), \\ \bar{\chi}^{\dot{\alpha}}(\mathbf{p}) &\rightarrow \bar{\chi}'^{\dot{\alpha}}(\mathbf{p}) = \bar{\chi}^{\dot{\beta}}(\Lambda^{-1}(A)\mathbf{p})(A^{-1})^*_{\dot{\beta}}{}^{\dot{\alpha}},\end{aligned}\tag{1.1}$$

where  $\Lambda(A)^\mu{}_\nu := \frac{1}{2} \text{Tr}(\bar{\sigma}^\mu A \sigma_\nu A^\dagger)$  is the  $A$ -induced four vector Lorentz transformation<sup>1</sup>. By introducing hermitian generators of the Lorentz group, the Lie algebra decomposes into 2 independent  $\mathfrak{su}(2)$  Lie algebras and hence can be characterized by two half integer numbers  $(j_1, j_2)$ . The  $\xi_\alpha \in (\frac{1}{2}, 0)$  is a left-handed Weyl spinor and  $\bar{\chi}^{\dot{\alpha}} \in (0, \frac{1}{2})$  is a right-handed Weyl spinor. Note, the dot over an index and/ or the bar over a function for objects transforming w.r.t.  $A^*$  instead of  $A$ .

$\sigma^\mu$  and  $\bar{\sigma}^\mu$  are defined with

$$(\sigma^\mu)_{\alpha\dot{\alpha}} := (\mathbb{1}_{\alpha\dot{\alpha}}, (\sigma^i)_{\alpha\dot{\alpha}})^\mu \quad \wedge \quad (\bar{\sigma}^\mu)^{\dot{\alpha}\alpha} := (\mathbb{1}^{\dot{\alpha}\alpha}, -(\sigma^i)^{\dot{\alpha}\alpha})^\mu,\tag{1.2}$$

and have the properties:

$$(\sigma^\mu)^\dagger = \sigma^\mu, \quad (\bar{\sigma}^\mu)^\dagger = \bar{\sigma}^\mu, \quad \text{Tr}(\sigma^\mu) = 0 = \text{Tr}(\bar{\sigma}^\mu),\tag{1.3}$$

$$\bar{\sigma}^\mu \sigma^\nu + \bar{\sigma}^\nu \sigma^\mu = 2g^{\mu\nu} \mathbb{1}, \quad \sigma^\mu \bar{\sigma}^\nu + \sigma^\nu \bar{\sigma}^\mu = 2g^{\mu\nu} \mathbb{1},\tag{1.4}$$

$$\text{Tr}(\bar{\sigma}^\mu \sigma^\nu) = \text{Tr}(\sigma^\mu \bar{\sigma}^\nu) = 2g^{\mu\nu}\tag{1.5}$$

$$(\sigma^\mu)_{\alpha\dot{\beta}} (\bar{\sigma}_\mu)^{\dot{\gamma}\delta} = 2\delta_\alpha^\delta \delta_{\dot{\beta}}^{\dot{\gamma}}, \quad (\sigma^\mu)^{\alpha\dot{\alpha}} (\sigma_\mu)^{\beta\dot{\beta}} = 2\varepsilon^{\alpha\beta} \varepsilon^{\dot{\alpha}\dot{\beta}}.\tag{1.6}$$

Dotted and undotted indices can be raised and lowered by "metric"-tensor

$$\begin{aligned}\varepsilon^{\alpha\beta} &:= (\varepsilon)^{\alpha\beta} = -\varepsilon^{\beta\alpha} \quad \wedge \quad \varepsilon^{\dot{\alpha}\dot{\beta}} := (\varepsilon)^{\dot{\alpha}\dot{\beta}}, \\ \varepsilon_{\alpha\beta} &:= (\varepsilon^{-1})_{\alpha\beta} \quad \wedge \quad \varepsilon_{\dot{\alpha}\dot{\beta}} := (\varepsilon^{-1})_{\dot{\alpha}\dot{\beta}}, \\ \delta_\beta^\alpha &= \varepsilon^{\alpha\gamma} \varepsilon_{\gamma\beta} \quad \wedge \quad \delta_{\dot{\beta}}^{\dot{\alpha}} = \varepsilon^{\dot{\alpha}\dot{\gamma}} \varepsilon_{\dot{\gamma}\dot{\beta}},\end{aligned}\tag{1.7}$$

which is invariant w.r.t  $\text{SL}(2, \mathbb{C})$ -transformation and defined by the matrix<sup>2</sup>

$$\varepsilon := +i\sigma^2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \varepsilon = \varepsilon^* = -\varepsilon^\dagger = -\varepsilon^T = -\varepsilon^{-1}.\tag{1.8}$$

<sup>3</sup> As can be checked by considering the explicit transformations (1.1), the following expressions transform in the correct way and establish thus the index raising and lowering:

$$\xi^\alpha = \varepsilon^{\alpha\beta} \xi_\beta, \quad \xi_\alpha = \varepsilon_{\alpha\beta} \xi^\beta, \quad \bar{\chi}^{\dot{\alpha}} = \varepsilon^{\dot{\alpha}\dot{\beta}} \bar{\chi}_{\dot{\beta}}, \quad \bar{\chi}_{\dot{\alpha}} = \varepsilon_{\dot{\alpha}\dot{\beta}} \bar{\chi}^{\dot{\beta}}\tag{1.9}$$

<sup>1</sup>Remember that  $\text{SL}(2, \mathbb{C})$  is the universal cover of the proper orthochronous Lorentz group  $L_+^\uparrow$  which is connected, i.e.  $\text{SL}(2, \mathbb{C})$  is simply-connected and homeomorphic to  $L_+^\uparrow$ .

<sup>2</sup>Our convention  $\varepsilon^{12} = +1 = -\varepsilon_{12}$  is the same as in [3, 7]. A change of sign would especially alter the signs in Equation 1.28.

<sup>3</sup>The Pauli matrices are defined as:  $\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ ,  $\sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$ ,  $\sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ , with  $(\sigma^i)^\dagger = \sigma^i$  and  $\sigma^i \sigma^j = \delta^{ij} + i\varepsilon^{ijk} \sigma^k$ .

Contraction of spinors are defined as

$$\begin{aligned}\xi_1 \xi_2 &:= (\xi_1)^\alpha (\xi_2)_\alpha = (\xi_1)_2 (\xi_2)_1 - (\xi_1)_1 (\xi_2)_2 = -(\xi_1)_\alpha (\xi_2)^\alpha, \\ \bar{\chi}_1 \bar{\chi}_2 &:= (\bar{\chi}_1)_{\dot{\alpha}} (\bar{\chi}_2)^{\dot{\alpha}} = (\bar{\chi}_1)^1 (\bar{\chi}_2)^{\dot{2}} - (\bar{\chi}_1)^{\dot{2}} (\bar{\chi}_2)^1.\end{aligned}\quad (1.10)$$

This definition automatically respects anti-commutation of fermions, e.g.  $\xi\phi = \xi^\alpha \phi_\alpha = -\phi_\alpha \xi^\alpha = \phi^\alpha \xi_\alpha = \phi\xi$ .

In addition,

$$\boxed{(\bar{\sigma}^\mu)^{\dot{\alpha}\alpha} = \varepsilon^{\alpha\beta} \varepsilon^{\dot{\alpha}\dot{\beta}} (\sigma^\mu)_{\beta\dot{\beta}} = (\sigma^\mu)^{\alpha\dot{\alpha}}}\quad (1.11)$$

can be shown. Define now:

$$(\sigma^{\mu\nu})_\alpha{}^\beta := \frac{i}{2} (\sigma^\mu \bar{\sigma}^\nu - \sigma^\nu \bar{\sigma}^\mu)_\alpha{}^\beta, \quad (\bar{\sigma}^{\mu\nu})^{\dot{\alpha}}{}_{\dot{\beta}} := \frac{i}{2} (\bar{\sigma}^\mu \sigma^\nu - \bar{\sigma}^\nu \sigma^\mu)^{\dot{\alpha}}{}_{\dot{\beta}} \quad (1.12)$$

and pay attention to the unusual index structure which yields for example:

$$(\sigma^{\mu\nu})^{\alpha\beta} := \frac{i}{2} \varepsilon^{\alpha\gamma} ((\sigma^\mu)_{\gamma\dot{\gamma}} (\bar{\sigma}^\nu)^{\dot{\gamma}\beta} - (\sigma^\nu)_{\gamma\dot{\gamma}} (\bar{\sigma}^\mu)^{\dot{\gamma}\beta}) \quad (1.13)$$

These matrices have the following properties:

$$\sigma^{\mu\nu} = -\sigma^{\nu\mu}, \quad \bar{\sigma}^{\mu\nu} = -\bar{\sigma}^{\nu\mu}, \quad \bar{\sigma}^{\mu\nu} = (\sigma^{\mu\nu})^\dagger, \quad \text{Tr}(\sigma^{\mu\nu}) = \text{Tr}(\bar{\sigma}^{\mu\nu}) = 0, \quad (1.14)$$

$$\sigma^\mu \bar{\sigma}^\nu = g^{\mu\nu} \mathbb{1} - i\sigma^{\mu\nu}, \quad \bar{\sigma}^\mu \sigma^\nu = g^{\mu\nu} \mathbb{1} - i\bar{\sigma}^{\mu\nu}, \quad (1.15)$$

$$\sigma^{0i} = -i\sigma^i, \quad \sigma^{ij} = \epsilon^{ijk} \sigma^k, \quad \bar{\sigma}^{0i} = i\sigma^i, \quad \bar{\sigma}^{ij} = \epsilon^{ijk} \sigma^k, \quad (1.16)$$

$$(\sigma^{\mu\nu})_{\alpha\beta} = (\sigma^{\mu\nu})_{\beta\alpha}, \quad (\bar{\sigma}^{\mu\nu})_{\dot{\alpha}\dot{\beta}} = (\bar{\sigma}^{\mu\nu})_{\dot{\beta}\dot{\alpha}}, \quad (1.17)$$

$$\begin{aligned}(\sigma^{\mu\nu})_\alpha{}^\beta (\sigma_{\mu\nu})_\gamma{}^\delta &= 4(\delta_\alpha^\delta \delta_\gamma^\beta + \varepsilon_{\alpha\gamma} \varepsilon^{\beta\delta}), \quad (\bar{\sigma}^{\mu\nu})^{\dot{\alpha}}{}_{\dot{\beta}} (\bar{\sigma}_{\mu\nu})^{\dot{\gamma}}{}_{\dot{\delta}} = 4(\delta_{\dot{\delta}}^{\dot{\gamma}} \delta_{\dot{\beta}}^{\dot{\alpha}} + \varepsilon^{\dot{\alpha}\dot{\gamma}} \varepsilon_{\dot{\beta}\dot{\delta}}), \\ (\sigma^{\mu\nu})_\alpha{}^\beta (\bar{\sigma}_{\mu\nu})^{\dot{\alpha}}{}_{\dot{\beta}} &= 0,\end{aligned}\quad (1.18)$$

$$\boxed{\Rightarrow (\sigma^{\mu\nu})^{\alpha_1\alpha_2} (\sigma_{\mu\nu})^{\alpha_3\alpha_4} = -4(\varepsilon^{\alpha_1\alpha_4} \varepsilon^{\alpha_2\alpha_3} + \varepsilon^{\alpha_1\alpha_3} \varepsilon^{\alpha_2\alpha_4})}\quad (1.19)$$

$$\frac{1}{2} \text{Tr}(\sigma^{\mu\nu} \sigma_{\rho\sigma}) \equiv \frac{1}{2} (\sigma^{\mu\nu})_\alpha{}^\beta (\sigma_{\rho\sigma})_\beta{}^\alpha = \delta_\rho^\mu \delta_\sigma^\nu - \delta_\sigma^\mu \delta_\rho^\nu - i\varepsilon^{\mu\nu}{}_{\rho\sigma}, \quad (1.20)$$

$$\frac{1}{2} \text{Tr}(\bar{\sigma}^{\mu\nu} \bar{\sigma}_{\rho\sigma}) \equiv \frac{1}{2} (\bar{\sigma}^{\mu\nu})^{\dot{\alpha}}{}_{\dot{\beta}} (\bar{\sigma}_{\rho\sigma})^{\dot{\beta}}{}_{\dot{\alpha}} = \delta_\rho^\mu \delta_\sigma^\nu - \delta_\sigma^\mu \delta_\rho^\nu + i\varepsilon^{\mu\nu}{}_{\rho\sigma}. \quad (1.21)$$

<sup>4</sup> <sup>5</sup>and are further self- and anti-self dual, i.e.:

$$\sigma^{\mu\nu} = \frac{i}{2} \varepsilon^{\mu\nu\rho\sigma} \sigma_{\rho\sigma}, \quad \bar{\sigma}^{\mu\nu} = -\frac{i}{2} \varepsilon^{\mu\nu\rho\sigma} \bar{\sigma}_{\rho\sigma}. \quad (1.22)$$

Not further that  $\sigma^{\mu\nu}/2$  and  $\bar{\sigma}^{\mu\nu}/2$  are generators of the (projective) representations of  $L_+^\uparrow$ :

$$\begin{aligned}\xi_\alpha(\mathbf{p}) &\rightarrow \xi'_\alpha(\mathbf{p}) = \left( e^{-\frac{i}{2} \omega_{\mu\nu} \frac{\sigma^{\mu\nu}}{2}} \right)_\alpha{}^\beta \xi_\beta(\Lambda^{-1}\mathbf{p}), \\ \bar{\chi}^{\dot{\alpha}}(\mathbf{p}) &\rightarrow \bar{\chi}'^{\dot{\alpha}}(\mathbf{p}) = \left( e^{-\frac{i}{2} \omega_{\mu\nu} \frac{\bar{\sigma}^{\mu\nu}}{2}} \right)^{\dot{\alpha}}{}_{\dot{\beta}} \bar{\chi}^{\dot{\beta}}(\Lambda^{-1}\mathbf{p}).\end{aligned}\quad (1.23)$$

<sup>4</sup>See [3, p. 59] or [1, p. 17–19] for a derivation.

<sup>5</sup>The four component epsilon tensor  $\varepsilon^{\mu\nu\rho\sigma}$  is chosen to fulfil  $\varepsilon^{0123} = -\varepsilon_{0123} = +1$ .

## 1.2 Dirac bi-spinor field

Let  $\xi_\alpha$  and  $\chi_\alpha$  be two component left-handed Weyl spinors (i.e.  $\xi_\alpha, \chi_\alpha \in (\frac{1}{2}, 0)$ ). Then, the hermitian conjugate  $(\chi^\alpha)^\dagger = \chi^{\dot{\alpha}}$  after raising an index transforms as a right-handed Weyl spinor (i.e.  $\chi^{\dot{\alpha}} \in (0, \frac{1}{2})$ ). Combination of these independent spinors yields the four component Dirac bi-spinor:

$$\psi(x) := \begin{pmatrix} \xi_\alpha(x) \\ (\chi^\dagger)^{\dot{\alpha}}(x) \end{pmatrix} \in \left(\frac{1}{2}, \frac{1}{2}\right). \quad (1.24)$$

Define now the Dirac matrices in the Weyl- or Chiral-representation:

$$\gamma^\mu := \begin{pmatrix} \mathbb{0}_{2 \times 2} & (\sigma^\mu)_{\alpha\dot{\beta}} \\ (\bar{\sigma}^\mu)^{\dot{\alpha}\beta} & \mathbb{0}_{2 \times 2} \end{pmatrix}, \quad \{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}. \quad (1.25)$$

Define also the adjoint Dirac-spinor  $\bar{\psi}$ :

$$\bar{\psi}(x) := (\chi^\alpha(x), (\xi^\dagger)_{\dot{\alpha}}(x)) = \psi^\dagger(x)\beta, \quad (1.26)$$

where  $\beta := \begin{pmatrix} 0 & \delta_{\dot{\beta}}^{\dot{\alpha}} \\ \delta_{\alpha}^{\beta} & 0 \end{pmatrix}$ , which is numerically identical to  $\gamma^0$ , but with a slightly different index structure. Lets define also the charge conjugated field

$$\psi^C(x) := \begin{pmatrix} \chi_\alpha(x) \\ (\xi^\dagger)^{\dot{\alpha}}(x) \end{pmatrix} = C(\bar{\psi})^T(x) \Rightarrow \bar{\psi}^C(x) = (\xi^\alpha(x), (\chi^\dagger)_{\dot{\alpha}}(x)) = -\psi^T(x)C^{-1}, \quad (1.27)$$

with the charge conjugation matrix

$$C := i\gamma^0\gamma^2 = \begin{pmatrix} \varepsilon_{\alpha\beta} & 0 \\ 0 & \varepsilon^{\dot{\alpha}\dot{\beta}} \end{pmatrix} = \begin{pmatrix} -\varepsilon^{\alpha\beta} & 0 \\ 0 & -\varepsilon_{\dot{\alpha}\dot{\beta}} \end{pmatrix}, \quad (1.28)$$

$C = C^* = -C^\dagger = -C^T = -C^{-1}$  and  $C^{-1}\gamma_\mu C = -\gamma_\mu^T$ . Finally, define

$$\sigma^{\mu\nu} := \frac{i}{2} [\gamma^\mu, \gamma^\nu] = \begin{pmatrix} (\sigma^{\mu\nu})_{\alpha}^{\beta} & 0 \\ 0 & (\bar{\sigma}^{\mu\nu})_{\dot{\beta}}^{\dot{\alpha}} \end{pmatrix}. \quad (1.29)$$

Combining all the previous results yields the following Dirac bi-spinor expressions:

$$\begin{aligned} \bar{\Psi}_1 \Psi_2 &= \chi_1^\alpha \xi_{2\alpha} + \xi_{1\dot{\alpha}}^\dagger \chi_2^{\dot{\alpha}}, \\ \bar{\Psi}_1 \gamma^\mu \Psi_2 &= \chi_1^\alpha (\sigma^\mu)_{\alpha\dot{\alpha}} \chi_2^{\dot{\alpha}} + \xi_{1\dot{\alpha}}^\dagger (\bar{\sigma}^\mu)^{\dot{\alpha}\alpha} \xi_{2\alpha}, \\ \bar{\Psi}_1 \sigma^{\mu\nu} \Psi_2 &= \chi_1^\alpha (\sigma^{\mu\nu})_{\alpha}^{\beta} \xi_{2\beta} + \xi_{1\dot{\alpha}}^\dagger (\bar{\sigma}^{\mu\nu})_{\dot{\beta}}^{\dot{\alpha}} \chi_2^{\dot{\beta}}, \\ \Psi_1^T C \Psi_2 &= \bar{\Psi}_1^C \Psi_2 = \xi_1^\alpha \xi_{2\alpha} + \chi_{1\dot{\alpha}}^\dagger \chi_2^{\dot{\alpha}}, \\ \Psi_1^T C \gamma^\mu \Psi_2 &= \xi_1^\alpha (\sigma^\mu)_{\alpha\dot{\alpha}} \chi_2^{\dot{\alpha}} + \chi_{1\dot{\alpha}}^\dagger (\bar{\sigma}^\mu)^{\dot{\alpha}\alpha} \xi_{2\alpha}, \\ \Psi_1^T C \sigma^{\mu\nu} \Psi_2 &= \xi_1^\alpha (\sigma^{\mu\nu})_{\alpha}^{\beta} \xi_{2\beta} + \chi_{1\dot{\alpha}}^\dagger (\bar{\sigma}^{\mu\nu})_{\dot{\beta}}^{\dot{\alpha}} \chi_2^{\dot{\beta}}, \\ \bar{\Psi}_1 C \bar{\Psi}_2^T &= \xi_{1\dot{\alpha}}^\dagger \xi_2^{\dot{\alpha}} + \chi_1^\alpha \chi_{2\alpha}, \\ \bar{\Psi}_1 \gamma^\mu C \bar{\Psi}_2^T &= \chi_1^\alpha (\sigma^\mu)_{\alpha\dot{\alpha}} \xi_2^{\dot{\alpha}} + \xi_{1\dot{\alpha}}^\dagger (\bar{\sigma}^\mu)^{\dot{\alpha}\alpha} \chi_{2\alpha}, \\ \bar{\Psi}_1 \sigma^{\mu\nu} C \bar{\Psi}_2^T &= \xi_{1\dot{\alpha}}^\dagger (\bar{\sigma}^{\mu\nu})_{\dot{\beta}}^{\dot{\alpha}} \xi_2^{\dot{\beta}} + \chi_1^\alpha (\sigma^{\mu\nu})_{\alpha}^{\beta} \chi_{2\beta}. \end{aligned} \quad (1.30)$$

**Notation of [3, 4]**

$$\boxed{\begin{aligned} q_L &= \begin{pmatrix} Q \\ 0 \end{pmatrix}, & u_R &= \begin{pmatrix} 0 \\ u_C^\dagger \end{pmatrix}, & d_R &= \begin{pmatrix} 0 \\ d_C^\dagger \end{pmatrix}, & l_L &= \begin{pmatrix} L \\ 0 \end{pmatrix}, & e_R &= \begin{pmatrix} 0 \\ e_C^\dagger \end{pmatrix}, \\ \bar{q}_L &= (0, Q^\dagger), & \bar{u}_R &= (u_C, 0), & \bar{d}_R &= (d_C, 0), & \bar{l}_L &= (0, L^\dagger), & \bar{e}_R &= (e_C, 0). \end{aligned}} \quad (1.31)$$

Note: In Benjamin Summ's example and parts of the program the fields are notated as follows:

$$\begin{aligned} q_L &= \mathbf{q} = \begin{pmatrix} Q \\ 0 \end{pmatrix}, & u_R &= \mathbf{u} = \begin{pmatrix} 0 \\ [\mathbf{u}_C] \end{pmatrix}, & d_R &= \mathbf{b} = \begin{pmatrix} 0 \\ [\mathbf{d}_C] \end{pmatrix}, \\ \bar{q}_L &= [\mathbf{q}] = (0, [Q]), & \bar{u}_R &= [\mathbf{u}] = ([\mathbf{u}_C], 0), & \bar{d}_R &= [\mathbf{b}] = ([\mathbf{d}_C], 0), \\ l_L &= \mathbf{l} = \begin{pmatrix} L \\ 0 \end{pmatrix}, & e_R &= \mathbf{e} = \begin{pmatrix} 0 \\ [\mathbf{e}_C] \end{pmatrix}, \\ \bar{l}_L &= [\mathbf{l}] = (0, [L]), & \bar{e}_R &= [\mathbf{e}] = ([\mathbf{e}_C], 0). \end{aligned} \quad (1.32)$$

Together with the above relations and Equation 1.30 consider some examples:

- $\bar{u}_R \gamma^\mu u_R = u_C \sigma^\mu u_C^\dagger$  with  $u_C^\alpha \equiv \chi^\alpha$  and  $(u_C^\dagger)^{\dot{\alpha}} \equiv (\chi^\dagger)^{\dot{\alpha}}$
- $\bar{e}_R l_L =: \bar{e} l = e_C L$
- $\bar{u}^T C d_R =: \bar{u}^C d = u_C^\dagger d_C^\dagger$ .

**1.3 Fieldstrength tensor**

Let's consider now the chiral basis of gauge bosons  $F_{L/R}$  and its relation to the hermitian  $F$  and  $\tilde{F}$  where  $\tilde{F}^{\mu\nu} = \frac{1}{2} \varepsilon^{\mu\nu\rho\sigma} F_{\rho\sigma}$ :

$$\begin{aligned} F_{L/R}^{\mu\nu} &= \frac{1}{2} (F^{\mu\nu} \pm i \tilde{F}^{\mu\nu}), \quad (F_L)^\dagger = F_R, \\ \Leftrightarrow F^{\mu\nu} &= F_L^{\mu\nu} + F_R^{\mu\nu}, \quad \tilde{F}^{\mu\nu} = i(F_R^{\mu\nu} - F_L^{\mu\nu}). \end{aligned} \quad (1.33)$$

Note also the (anti-)self-duality of  $F_{R/L}$ , i.e.  $F_{L/R}^{\mu\nu} = \pm \frac{i}{2} \varepsilon^{\mu\nu\rho\sigma} F_{\rho\sigma}^{L/R}$ .

Since

$$\begin{aligned} F_L^{\mu\nu} (\sigma_{\mu\nu})_\alpha^\beta &= F^{\mu\nu} (\sigma_{\mu\nu})_\alpha^\beta = i \tilde{F}^{\mu\nu} (\sigma_{\mu\nu})_\alpha^\beta, & F_L^{\mu\nu} (\bar{\sigma}_{\mu\nu})^{\dot{\alpha}}_{\dot{\beta}} &= 0, \\ F_R^{\mu\nu} (\bar{\sigma}_{\mu\nu})^{\dot{\alpha}}_{\dot{\beta}} &= F^{\mu\nu} (\bar{\sigma}_{\mu\nu})^{\dot{\alpha}}_{\dot{\beta}} = -i \tilde{F}^{\mu\nu} (\bar{\sigma}_{\mu\nu})^{\dot{\alpha}}_{\dot{\beta}}, & F_R^{\mu\nu} (\sigma_{\mu\nu})_\alpha^\beta &= 0, \end{aligned} \quad (1.34)$$

it seems reasonable to define

$$\begin{aligned} F_{L\beta\alpha} &= F_{L\alpha\beta} := \frac{i}{2} F_{\mu\nu} \sigma_{\alpha\beta}^{\mu\nu} \in (1, 0) \text{ and } F_{R\dot{\beta}\dot{\alpha}} = F_{R\dot{\alpha}\dot{\beta}} := -\frac{i}{2} F_{\mu\nu} \bar{\sigma}_{\dot{\alpha}\dot{\beta}}^{\mu\nu} \in (0, 1) \\ \Rightarrow (F_{L\alpha\beta})^\dagger &= F_{R\dot{\alpha}\dot{\beta}}. \end{aligned} \quad (1.35)$$

Due to Equation 1.21 the inverse is given by

$$\boxed{\begin{aligned} F_{\mu\nu} &= -\frac{i}{4} \left( F_R^{\dot{\alpha}\dot{\beta}} (\bar{\sigma}_{\mu\nu})_{\dot{\alpha}\dot{\beta}} - F_{L\alpha\beta} \sigma_{\mu\nu}^{\alpha\beta} \right), \\ \tilde{F}_{\mu\nu} &= -\frac{1}{4} \left( F_R^{\dot{\alpha}\dot{\beta}} (\bar{\sigma}_{\mu\nu})_{\dot{\alpha}\dot{\beta}} + F_{L\alpha\beta} \sigma_{\mu\nu}^{\alpha\beta} \right). \end{aligned}} \quad (1.36)$$

### 1.4 (Covariant) Derivative

$$D_{\alpha\dot{\alpha}} := D_{\mu}\sigma_{\alpha\dot{\alpha}}^{\mu} \in (\frac{1}{2}, \frac{1}{2}) \text{ and } D_{\dot{\alpha}\alpha} := D_{\mu}\bar{\sigma}_{\dot{\alpha}\alpha}^{\mu} \stackrel{(1.11)}{=} D_{\mu}\sigma_{\alpha\dot{\alpha}}^{\mu} = D_{\alpha\dot{\alpha}} \Rightarrow D_{\alpha\dot{\alpha}} = D_{\dot{\alpha}\alpha} \quad (1.37)$$

$$\Leftrightarrow \boxed{D_{\mu} = \frac{1}{2}D_{\alpha\dot{\alpha}}(\bar{\sigma}_{\mu})^{\dot{\alpha}\alpha} = \frac{1}{2}D_{\dot{\alpha}\alpha}(\sigma_{\mu})^{\alpha\dot{\alpha}} = -\frac{1}{2}D_{\alpha}^{\dot{\alpha}}(\sigma_{\mu})_{\dot{\alpha}}^{\alpha}} \quad (1.38)$$

Index and derivative structure has to be understood as:

$$D\psi_{\alpha\beta\dot{\beta}}\psi_{\gamma} = (D\psi_{\alpha})_{\beta\dot{\beta}}\psi_{\gamma}. \quad (1.39)$$

Note further:

$$D_{\mu}D^{\mu} = \frac{1}{2}D^{\dot{\alpha}\alpha}D_{\alpha\dot{\alpha}}. \quad (1.40)$$

### 1.5 Adjoint Fields

First, some group theory stuff [s. 6, S. 482–488]:

$$[T^A, T^B] = if^{ABC}T^C, \quad \text{Tr}(T^AT^B) = T_{ji}^AT_{ij}^B = T_F\delta^{AB}, \quad \sum_A (T^AT^A)_{ij} = C_F\delta_{ij}, \quad (1.41)$$

$$f^{ACD}f^{BCD} = C_A\delta^{AB}, \quad f^{ABC} = -\frac{i}{T_F} \text{Tr}([T^A, T^B]T^C), \quad (1.42)$$

$$\boxed{\sum_A T_{ij}^AT_{kl}^A = T_F \left( \delta_{il}\delta_{kj} - \frac{1}{d_F}\delta_{ij}\delta_{kl} \right)}, \quad (1.43)$$

where  $T_F = \frac{1}{2}$ ,  $C_A = N = d_F$  and  $C_F = \frac{N^2-1}{2N}$  for  $\text{SU}(N)$ .  $(T^A)^b_c := \frac{(\lambda^A)^b_c}{2}$  for  $\text{SU}(3)$  with Gell-Mann matrices  $\lambda^A$  and  $(T^I)^j_k := \frac{(\tau^I)^j_k}{2}$  for  $\text{SU}(2)$ , with Pauli matrices  $\tau^I$ . Note further that the structure constants  $f^{ABC}$  are totally antisymmetric and (1.43) is essentially the Fierz identity for  $\text{SU}(N)$ .

$$\boxed{\begin{aligned} \varepsilon_{acd}(\lambda^A)^d_b G^A &= G_{abc} \sim \boxed{\begin{smallmatrix} a & b \\ c \end{smallmatrix}} \Leftrightarrow G^A = \frac{(\lambda^A)^b_c}{4} \varepsilon^{ace} G_{abc}, \\ \varepsilon_{abc}(Q^{\dagger})^c &= Q_{ab}^{\dagger} \sim \boxed{\begin{smallmatrix} a \\ b \end{smallmatrix}} \Leftrightarrow (Q^{\dagger})^c = \frac{1}{2} \varepsilon^{abc} Q_{bc}^{\dagger}, \\ \varepsilon_{jk}(\tau^I)^k_i W^I &= W_{(ij)} \sim \boxed{\begin{smallmatrix} i & j \end{smallmatrix}} \Leftrightarrow W^I = \frac{(\tau^I)^i_j}{2} \varepsilon^{jk} W_{(ik)}, \\ \varepsilon_{ij}(H^{\dagger})^j &= H_i \sim \boxed{i} \Leftrightarrow (H^{\dagger})^i = \varepsilon^{ij} H_j, \end{aligned}} \quad (1.44)$$

where  $W_{(ik)} := (W_{ik} + W_{ki})/2$  is the symmetric part of  $W$ . Note that also  $u_{\mathbb{C}}$  and  $d_{\mathbb{C}}$  are transforming in the anti-fundamental representation of  $\text{SU}(3)$  and are rewritten similar as  $Q^{\dagger}$ . Further, also parts of the lepton and quark doublet,  $L^{\dagger}$ ,  $Q^{\dagger}$  and  $e_{\mathbb{C}}$  are rewritten like the Higgs field  $H$ .

### 1.6 Examples from Benjamin Summ's direct output

The goal of this section is to examine some examples of Benjamin Summ's output and convert them in our Notation and Operator Basis. In the beginning, the operators are converted to the correct notation, where

- no fields are directly contracted with other fields,
- only fundamental indices occur,
- "smallest" index, w.r.t. the number on the index should be the first one in each  $\varepsilon$ ,
- fields are sorted with increasing helicity.

This rewriting allows to write each operator as linear combination  $\mathcal{O} = T^{abcdef} \Theta_{abcdef}$ .

Let's consider the 58th Term in Benjamin's output and (1.38), (1.44) and (1.31), where  $l_L := \begin{pmatrix} (L)_\alpha \\ 0 \end{pmatrix}$ :

$$\begin{aligned}
H^{\dagger,j} H_i (\bar{\psi}_L)_{i,r} (y_e y_e^\dagger)_{rt} (l_L)_{j,t} &= H^{\dagger,j} H_i D_\mu (\bar{l}_L)_i \gamma^\mu (y_e y_e^\dagger) (l_L)_j \\
&= \varepsilon^{jk} H_k^\dagger H_i (D_\mu (L^\dagger)_{i,\dot{\alpha}} (\bar{\sigma}^\mu)^{\dot{\alpha}\alpha} (y_e y_e^\dagger) L_{j\alpha} \\
&= \varepsilon^{\alpha\beta} \varepsilon_{\dot{\alpha}\dot{\beta}} \varepsilon^{jk} H_k^\dagger H_i (D (L^\dagger)_i)_{\dot{\beta}}^{\dot{\alpha}} (y_e y_e^\dagger) L_{j\alpha} \\
&= \varepsilon^{jk} \varepsilon^{\alpha\beta} \varepsilon_{\dot{\alpha}\dot{\beta}} (y_e y_e^\dagger)_{rt} H_k^\dagger H_i (D (L^\dagger)_i)^r_{\dot{\beta}}^{\dot{\alpha}} L_{j\alpha}^t
\end{aligned} \tag{1.45}$$

From step two to three (1.6) is used. In a next step, the 35th term is inspected:

$$\begin{aligned}
(T^I T^J)_i^j H^{\dagger,i} H_j V_{\mu\nu}^I V^{J,\mu\nu} &= -\frac{1}{16} \left( \varepsilon^{\alpha_1\beta_2} \varepsilon^{\alpha_2\beta_1} + \varepsilon^{\alpha_1\alpha_2} \varepsilon^{\beta_2\beta_1} \right) (T^I T^J)_i^j \varepsilon^{ik} H_k H_j V_{L\alpha_1\beta_1}^I V_{L\alpha_2\beta_2}^J \\
&\quad - \frac{1}{16} \left( \varepsilon_{\dot{\alpha}_1\dot{\beta}_2} \varepsilon_{\dot{\alpha}_2\dot{\beta}_1} + \varepsilon_{\dot{\alpha}_1\dot{\alpha}_2} \varepsilon_{\dot{\beta}_2\dot{\beta}_1} \right) (T^I T^J)_i^j \varepsilon^{ik} H_k H_j V_R^{I,\dot{\alpha}_1\dot{\beta}_1} V_R^{J,\dot{\alpha}_2\dot{\beta}_2},
\end{aligned} \tag{1.46}$$

where we used to write the Field strength tensors in the chiral notation. Next, the SU(2) part is rewritten and fully expressed by fundamental indices, with the help of (1.43):

$$\begin{aligned}
\varepsilon^{ik} (T^I T^J)_i^j V^I V^J &= -\varepsilon^{ki} \left( (T^I)_{ik} (T^I)^l_m \right) \left( (T^J)^{kj} (T^J)^p_q \right) \varepsilon^{mn} \varepsilon^{qr} V_{(ln)} V_{(pr)} \\
&\vdots \\
&= -\varepsilon^{ki} \varepsilon_i^n \varepsilon^{lr} \delta^{pj} V_{(ln)} V_{(pr)} \\
&= -\varepsilon^{ki} \varepsilon^{in} \varepsilon^{lr} \varepsilon^{pq} \varepsilon^{qj} V_{(ln)} V_{(pr)}.
\end{aligned} \tag{1.47}$$

This yields for the whole expression

$$\begin{aligned}
(T^I T^J)_i^j H^{\dagger,i} H_j V_{\mu\nu}^I V^{J,\mu\nu} &= -\frac{1}{16} \left( \varepsilon^{\alpha_1\beta_2} \varepsilon^{\beta_1\alpha_2} + \varepsilon^{\alpha_1\alpha_2} \varepsilon^{\beta_1\beta_2} \right) \varepsilon^{ip} \varepsilon^{pl} \varepsilon^{kn} \varepsilon^{mq} \varepsilon^{qj} H_i^\dagger H_j V_{L\alpha_1\beta_1,(kl)} V_{L\alpha_2\beta_2,(mn)} \\
&\quad - \frac{1}{16} \left( \varepsilon_{\dot{\alpha}_1\dot{\beta}_2} \varepsilon_{\dot{\beta}_1\dot{\alpha}_2} + \varepsilon_{\dot{\alpha}_1\dot{\alpha}_2} \varepsilon_{\dot{\beta}_1\dot{\beta}_2} \right) \varepsilon^{ip} \varepsilon^{pl} \varepsilon^{kn} \varepsilon^{mq} \varepsilon^{qj} H_i^\dagger H_j V_{R,\dot{\alpha}_1\dot{\beta}_1,(kl)} V_{R,\dot{\alpha}_2\dot{\beta}_2,(mn)}.
\end{aligned} \tag{1.48}$$

For term 103 one get's:

$$\begin{aligned}
&H_i H_k H^{\dagger,j} \varepsilon_{ik} \varepsilon_{lj} (y_u y_u^\dagger)_{tr} (\bar{q}_L)_{l,t}^a (d_R)_r^a \\
&= \varepsilon_{ik} \varepsilon_{lj} \varepsilon_{jm} (y_u y_u^\dagger)_{tr} H_i H_k H_m^\dagger (0, (Q^\dagger)_{l,t,\dot{\alpha}}^a) \cdot \begin{pmatrix} 0 \\ (d_\mathbb{C}^\dagger)_r^{a,\dot{\alpha}} \end{pmatrix} \\
&= \frac{1}{4} (\delta^{bd} \delta^{ce} - \delta^{be} \delta^{cd}) \varepsilon_{ik} \varepsilon_{lj} \varepsilon_{jm} (y_u y_u^\dagger)_{tr} \varepsilon_{\dot{\alpha}\dot{\beta}} H_i H_k H_m^\dagger (Q^\dagger)_{l,t}^{bc,\dot{\beta}} (d_\mathbb{C}^\dagger)_r^{de,\dot{\alpha}}
\end{aligned} \tag{1.49}$$

**Lorentz group** Unfortunately, due to the structure of the Lorentz group all known redundancy relations have to be implemented manually. These relations are:

1. Derivatives should only act on the first field. If this isn't the case integration by parts and the Leibniz rule are used to rewrite them. This has to be done, because otherwise the corresponding YT of the auxiliary  $SU(N)$  group aren't SSYT:  $(D^{r_1-|h_1|}\Phi_1)\dots \simeq -\Phi_1(D^{r_1-|h_1|}\dots)$ ,
2. Spinor indices of derivatives contracted with indices from the first two building blocks should be rewritten:  $\Phi_{1,\alpha\dots}(D_\alpha^\alpha D^{n_2-1}\Phi_2)\dots \simeq -\Phi_{1,\alpha\dots}(D^{n_2-1}\Phi_2)\dots(D_\alpha^\alpha\dots)\Phi_{1,\alpha\dots}\Phi_2^{\dot{\alpha}\dots}(D_\alpha^\alpha D^{n_2-1}\Phi_3) - \Phi_{1,\alpha\dots}\Phi_2^{\dot{\alpha}\dots}(D^{n_2-1}\Phi_3)\dots(D_\alpha^\alpha\dots)$ ,
3. Two derivatives acting on the second and third field are substituted by:  $\Phi_1 D^2(\Phi_2\Phi_3\dots) = \Phi_1(D^2\Phi_2)\Phi_3\dots + \Phi_1\Phi_2(D^2\Phi_3)\dots + \Phi_1\Phi_2\Phi_3(D^2\dots) + 2\Phi_1(D\Phi_2)(D\Phi_3)\dots + 2\Phi_1(D\Phi_2)\Phi_3(D\dots) + 2\Phi_1\Phi_2(D\Phi_3)(D\dots)$ ,
4. Schouten identities have to be implemented manually by substituting  $\varepsilon_{\alpha_i\alpha_l}\varepsilon_{\alpha_j\alpha_k} = -\varepsilon_{\alpha_i\alpha_j}\varepsilon_{\alpha_k\alpha_l} - \varepsilon_{\alpha_i\alpha_k}\varepsilon_{\alpha_l\alpha_j}$   
 $\tilde{\varepsilon}_{\dot{\alpha}_i\dot{\alpha}_l}\tilde{\varepsilon}_{\dot{\alpha}_j\dot{\alpha}_k} = -\tilde{\varepsilon}_{\dot{\alpha}_i\dot{\alpha}_j}\tilde{\varepsilon}_{\dot{\alpha}_k\dot{\alpha}_l} - \tilde{\varepsilon}_{\dot{\alpha}_i\dot{\alpha}_k}\tilde{\varepsilon}_{\dot{\alpha}_l\dot{\alpha}_j}$ .

Applying of all the relations brings operators in the form:  $\mathcal{M} = \sum_i m_i \mathcal{M}_i^{(y)}$ , where  $\mathcal{M}^{(y)}$  consist of fields of the same type only (s. e.g. (1.51)).

The only relation relevant for the above considered examples is the last one and yields for the 35th term, (1.48):

$$\begin{aligned} (T^I T^J)_i^j H^{\dagger,i} H_j V_{\mu\nu}^I V^{J,\mu\nu} = & -\frac{1}{16} \left( 2\varepsilon^{\alpha_1\alpha_2}\varepsilon^{\beta_1\beta_2} - \varepsilon^{\alpha_1\beta_1}\varepsilon^{\alpha_2\beta_2} \right) \varepsilon^{ip}\varepsilon^{pl}\varepsilon^{kn}\varepsilon^{mq}\varepsilon^{qj} H_i H_j V_{L\alpha_1\beta_1,(kl)} V_{L\alpha_2\beta_2,(mn)} \\ & -\frac{1}{16} \left( 2\varepsilon^{\dot{\alpha}_1\dot{\alpha}_2}\varepsilon^{\dot{\beta}_1\dot{\beta}_2} - \varepsilon^{\dot{\alpha}_1\dot{\beta}_1}\varepsilon^{\dot{\alpha}_2\dot{\beta}_2} \right) \varepsilon^{ip}\varepsilon^{pl}\varepsilon^{kn}\varepsilon^{mq}\varepsilon^{qj} H_i H_j V_{R,(kl)}^{\dot{\alpha}_1\dot{\beta}_1} V_{R,(mn)}^{\dot{\alpha}_2\dot{\beta}_2}. \end{aligned} \quad (1.50)$$

Identify the following tensors

$$\begin{aligned} \mathcal{M}^{(y)}_1 &= H_i H_j V_{L\alpha_1\beta_1,(kl)} V_{L\alpha_2\beta_2,(mn)}, \\ \mathcal{M}^{(y)}_2 &= H_i H_j V_{R,(kl)}^{\dot{\alpha}_1\dot{\beta}_1} V_{R,(mn)}^{\dot{\alpha}_2\dot{\beta}_2}, \end{aligned} \quad T_{SU(2)}^{(y)}_1 = \varepsilon^{ip}\varepsilon^{pl}\varepsilon^{kn}\varepsilon^{mq}\varepsilon^{qj}. \quad (1.51)$$

**Gauge Groups** Express now the gauge group structures which occurred above (e.g. (1.51)) and are part of the singlet representation of the gauge group by basis tensors of the singlet basis for the gauge group we already know. This is essentially simplified by the fact, that it is possible to define a scalar (inner) product between two gauge group tensors  $T_1$  and  $T_2$ :

$$(T_1, T_2) := \sum_{a_1, a_2, \dots} T_1^{a_1 a_2 \dots} T_2^{a_1 a_2 \dots}. \quad (1.52)$$

The singlet basis vectors are denoted by  $\{e_i := (T_G^{\tilde{y}})_i\}$  and are not necessarily orthogonal. Therefore, the introduction of the symmetric Gram-matrix  $G_{ij} := (e_i, e_j)$  consisting all possible scalar products of the basis tensors is necessary and after defining the tensor  $v_i := (G_{ij})^{-1}e_j$  allows to express an arbitrary tensor in the singlet- or y-basis:

$$T_G = \sum_i (v_i, T_G) e_i, \quad (1.53)$$

where the Gram-matrix ensures normalization  $(e_i, v_i) = 1$ .

Combination of the Lorentz group decomposition relations and projections in the gauge group sector allows to write every operator in the y-basis as

$$\mathcal{O} = T_G \mathcal{M} = \sum_{i=1}^{\mathcal{N}_G} t_i T_G^{\tilde{y}i} \sum_{j=1}^{\mathcal{N}_M} m_j \mathcal{M}_j^{(y)} =: \sum_{i,j} t_i m_j \mathcal{O}_{i,j}^{(y)}. \quad (1.54)$$

$\mathcal{N}_G$  and  $\mathcal{N}_M$  denote the dimension of the spaces spanned by  $T_G$  and  $\mathcal{M}$ .

**Redundant operators** The operators of Benjamin Summ, now in the correct notation, do not necessarily form a basis, but the operators of the y-basis do. Thus, it seems reasonable to extract an linearly independent set of operators from the BS operators. The procedure can be described as follows:

1. Search for operators (in BS's output) build out of the same field.
2. Express every operator in the y-Basis as described above, so that all operators have the form<sup>6</sup>  $\mathcal{O} = \kappa_i \mathcal{O}_i^{(y)}$ .
3. Construct a matrix out of all coefficients: 
$$\begin{pmatrix} \mathcal{O} \\ \mathcal{O}' \\ \vdots \end{pmatrix} = \begin{pmatrix} \kappa_1 & \kappa_2, \dots & \\ \kappa'_1 & \kappa'_2 & \dots \\ \vdots & & \ddots \end{pmatrix} \begin{pmatrix} \mathcal{O}_1^{(y)} \\ \mathcal{O}_2^{(y)} \\ \vdots \end{pmatrix} =: K_{ij} \mathcal{O}_i^{(y)}$$
4. Determine rank  $r := \text{rank}(K_{ij})$  and choose  $r$  linearly independent rows of the matrix  $K_{ij}$  and build from them the new matrix  $\bar{K}_{ij}$ .

This gives the so called m-basis, which is defined by the  $r$  linearly independent operators:

$$\mathcal{O}_i^{(m)} = \sum_{j=1}^{\mathcal{N}} \bar{K}_{ij} \mathcal{O}_j^{(y)}, \quad i = 1, \dots, r. \quad (1.55)$$

The operators in the m-basis span the whole space of the y-basis if and only if  $r = \mathcal{N}$ .

## 1.7 Equation of Motion complication

### 1.8 Real singlet scalar extension of the SM (SSM)

The electroweak (EW) part of the SM is given by the Lagrangian:

$$\begin{aligned} \mathcal{L}_{\text{SM}} = & (D_\mu H)^\dagger (D^\mu H) + \mu_h^2 H^\dagger H - \frac{1}{2} \lambda_h (H^\dagger H)^2 - \frac{1}{4} B_{\mu\nu} B^{\mu\nu} - \frac{1}{4} W_{\mu\nu}^a W^{a\mu\nu} \\ & + \sum_{f=q,u,d,\ell,e} \bar{f} i \not{D} f - \left( y_u \bar{q} \tilde{H} u + y_d \bar{q} H d + y_e \bar{\ell} H e + \text{h.c.} \right), \end{aligned} \quad (1.56)$$

where  $H$  denotes the SM Higgs doublet and  $\tilde{H}_i := (i\sigma^2 H^*)_i = -\varepsilon_{ij} (H_j)^*$  denotes the charge-conjugated Higgs field. Note further that right-handed neutrino fields are not considered in this investigation and the sum over flavour indices is implicit. The covariant derivative in the only EW case (QCD is ignored, since it is irrelevant for the considered matching) is specified as

$$D_\mu = \partial_\mu - ig_1 Y B_\mu - ig_2 \frac{\sigma^a}{2} W_\mu^a. \quad (1.57)$$

$g_1$  and  $B_\mu$  are the  $U(1)_Y$ ,  $g_2$  and  $W_\mu^a$  are the  $SU(2)_L$  gauge coupling and gauge fields. The Yukawa couplings are  $y_u$ ,  $y_d$  and  $y_e$ .  $q$  and  $\ell$  denote the left-handed quark and lepton doublets, whereas  $u$ ,  $d$  and  $e$  are the right-handed fermion singlets. The Lagrangian of the real scalar singlet is restricted by Lorentz and gauge invariance and is given by

$$\mathcal{L}_\phi = \frac{1}{2} (\partial_\mu \phi)^2 - \frac{1}{2} M^2 \phi^2 - A H^\dagger H \phi - \frac{1}{2} \kappa H^\dagger H \phi^2 - \frac{1}{3!} \mu \phi^3 - \frac{1}{4!} \lambda_\phi \phi^4. \quad (1.58)$$

The parameters which don't have mass dimension zero are parametrized by the scalar field mass  $M$ :

$$A = aM, \quad \mu = \hat{\mu}M, \quad \mu_h = \hat{\mu}_h M, \quad \mu_M = \hat{\mu}_M M. \quad (1.59)$$

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<sup>6</sup>For ease of notation an general index  $i := (i, j)$  was introduced, which yield for (1.54):  $\mathcal{O} = t_i m_j \mathcal{O}_{i,j}^{(y)} \Leftrightarrow \mathcal{O} = \kappa_i \mathcal{O}_i^{(y)}$

In the BSUOLEA output the constants are specified as follows:

$$d = \text{d}, \quad \varepsilon = \text{eps}, \quad \mu_h = \text{Mu}, \quad \lambda_h = \text{lambdah}, \quad g_1 = \text{g1}, \quad g_2 = \text{g2}, \quad g_3 = \text{g3}, \quad (1.60)$$

$$M = \text{Ms}, \quad A = \text{At}, \quad \kappa = \text{kappa}, \quad \mu = \text{mu}, \quad \lambda_\phi = \text{lambdaphi}, \quad \mu_M = \backslash[\text{Mu}], \quad (1.61)$$

where  $\varepsilon$  denote the epsilon of dimensional regularization (DREG) in  $d$  dimensions, i.e.  $d = 4 - \varepsilon$ <sup>7</sup>. The python program leaves most of the names untouched, but  $\mu_M = \text{muM}$  has to be renamed for FORM compatibility. Moreover, the logarithm  $\ln \frac{M_s^2}{\mu_M^2} = [2\text{L}[\text{Ms}, \text{muM}]]$  is rewritten.

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<sup>7</sup>This is the at least the definition of the DREG- $\varepsilon$  in [8, p. 57]. Note, that the DREG- $\varepsilon$  in [2, p. 4] is defined with  $d = 4 - 2\varepsilon$ .